

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“Jnana Sangama”, Belagavi – 590018, Karnataka.



A Mini-Project Report
on

“Earthquake Monitoring and Alerting System”

Submitted in partial fulfilment for the award of degree of

BACHELOR OF ENGINEERING

in

ELECTRONICS & COMMUNICATION ENGINEERING

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for the Academic Year 2025-26

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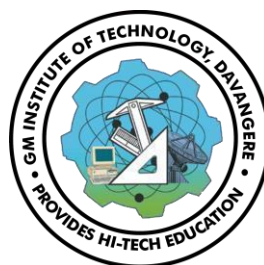
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ABSTRACT

This project proposes a low-cost and reliable Earthquake Monitoring and Alerting System built using an Arduino microcontroller and a GSM module to deliver rapid seismic warnings in real time. The system integrates a vibration sensor or accelerometer (such as an MPU6050 or SW-420) to continuously measure ground motion and detect the presence of unusual tremors. The Arduino processes incoming sensor data using threshold-based or calibrated sensitivity algorithms to distinguish between normal environmental vibrations and potential earthquake activity. When seismic activity exceeding the predefined threshold is detected, the Arduino immediately activates the GSM module to transmit SMS alerts to predefined contacts, including residents, disaster response teams, and local authorities. The system's architecture emphasizes simplicity, portability, and affordability, making it viable for deployment in rural, isolated, or economically constrained regions where advanced seismic monitoring infrastructure is not available. Its capability to function independently without internet connectivity ensures dependable operation even during network outages or disaster conditions. Additionally, the device supports real-time monitoring, rapid notification, and expandability for integration with sirens, IoT platforms, or cloud-based dashboards.

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Chapter 1

Introduction

Earthquakes are natural disasters that occur without warning and can cause severe damage to life and property. Due to the unpredictable nature of earthquakes, it is important to have systems that can detect vibrations early and alert people immediately. An Earthquake Monitoring and Alerting System helps in identifying unusual ground movements and provides quick warnings to minimize risks. In this project, an Arduino Uno microcontroller is used as the central processing unit. It receives inputs from an SW-420 vibration sensor, which is capable of detecting even small vibrations in the environment. When the sensor detects a vibration level beyond the set threshold, it sends a signal to the Arduino, which processes the data and activates an alarm system.

To alert users, a buzzer produces a loud sound indicating danger. Additionally, LED indicators are used to visually show the vibration intensity—green for normal conditions, yellow for minor vibrations, and red for high vibrations or earthquake-like activity. An LCD display continuously shows real-time vibration readings and the current status of the system, such as “Normal,” “Vibration Detected,” or “Earthquake Alert.”

Earthquake Monitoring System - Event Distribution

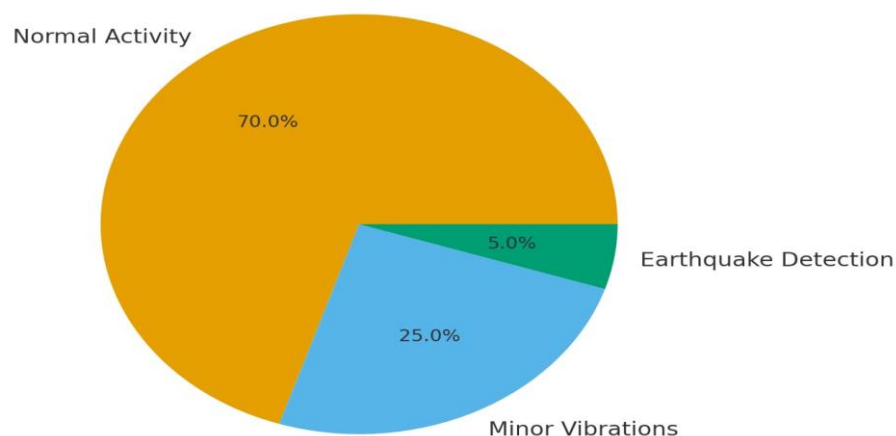


Figure 1: Earthquake Monitoring and Alerting System

1.1 Scope of the Project

The scope of this project extends to building a functional and reliable earthquake monitoring and alerting system that uses vibration-sensing technology to identify abnormal ground movements. Using the Arduino Uno as the controller, the system continuously monitors input from the SW-420 vibration sensor and reacts instantly when vibration levels exceed the predefined safety threshold. The scope includes integrating multiple alerting methods such as a buzzer for audible alerts and LEDs for visual indication of vibration intensity. An LCD module is also included to provide real-time feedback, making the system user-friendly and informative. This project is designed primarily for small-scale environments like homes, classrooms, college labs, small offices, and local communities where a simple, low-cost earthquake alert system is beneficial. It also serves as an educational tool to help students understand sensors, microcontrollers, and disaster management technologies.

Although the system does not aim to predict earthquakes with scientific accuracy, it effectively demonstrates the practical application of embedded systems in early warning technology. Additionally, the scope opens opportunities for future expansion, such as adding wireless communication, IoT-based mobile alerts, automatic emergency shutdowns, data logging, and integration with community alarm networks to make the solution more advanced and practical.

1.2 Problem Statement

Earthquakes are unpredictable natural disasters that often strike without any prior warning, causing widespread destruction, structural damage, and loss of life. In many regions, especially in developing areas, access to advanced and expensive earthquake detection systems is limited. As a result, households, schools, small offices, and community buildings remain highly vulnerable during sudden vibrations or tremors. Traditional early warning systems are complex, costly, and require professional installation, making them unsuitable for small-scale use. There is a need for an affordable, easy-to-use, and reliable system that can continuously monitor ground vibrations and alert individuals instantly when abnormal activity is detected. The problem becomes more serious in places where people cannot visually sense early vibrations or wake up during mild tremors. To address this challenge, the project proposes the development of a simple Arduino-based earthquake monitoring and alerting system using components such as the SW-420 vibration sensor, buzzer, LEDs, and an LCD display.

1.3 Objectives

The objectives of the mini-project work are as follows:

- To detect ground vibrations using the SW-420 vibration sensor accurately.
- To monitor vibration levels continuously with the help of Arduino Uno.
- To provide immediate alerts during abnormal or high-intensity vibrations using a buzzer and LEDs.
- To display real-time vibration status and system messages on the LCD screen.

1.4 Motivation

The motivation behind developing an earthquake monitoring and alerting system comes from the growing need for safety and preparedness in areas vulnerable to seismic activity. Earthquakes often occur without warning, leaving people with very little time to react. Many homes, schools, and small buildings lack affordable early-warning devices, increasing the risk of injury, panic, and property damage during sudden tremors. This situation creates a strong need for a simple, low-cost technology that can detect vibrations early and alert people immediately. Using basic components like the Arduino Uno, SW-420 vibration sensor, buzzer, LEDs, and LCD display makes it possible to design an easy-to-build system that can be used by anyone even without technical expertise. The project is also motivated by the desire to apply embedded system concepts to real-life problems, demonstrating how small electronic devices can contribute to public safety. Additionally, this project encourages awareness about disaster management and empowers communities to take preventive measures. The motivation also includes learning engineering skills such as sensor interfacing, real-time monitoring, and alarm systems, while contributing to a solution that can help protect lives during emergencies.

1.5 Organization of the Report

Chapter 1: Gives the introduction about the project “Earthquake Monitoring and Alerting System”. It is also brief about the Scope of the project, Problem Statement, Objectives and Motivation.

Chapter 2: Provides the Literature Review of the various research papers.

Chapter 3: This chapter briefly explains about project methodology, block diagram and flowchart.

Chapter 4: Provides information about the Hardware and software requirements.

Chapter 5: Provides information about the project implementation.

Chapter 6: Explains the results and discussion.

Chapter 7: Gives information about the Advantages, Disadvantages, and Applications of the Mini-project entitled “Earthquake Monitoring and Alerting System”.

Chapter 8: This chapter presents the conclusion and provides the references for the project.

Chapter 2

Literature Review

Under the literature review, various research papers, journals, and technical reports related to earthquake detection, vibration sensing, early-warning technologies, and embedded alerting systems were studied. These works reveal the advancements in low-cost seismic monitoring using sensors like SW-420, MEMS accelerometers, Arduino controllers, GSM modules, and IoT platforms. The surveyed literature helps in understanding existing solutions, their limitations, and the motivation behind developing a simple real-time earthquake alert system.

[1] Title: “HOME LEVEL EARTHQUAKE ALERTING DEVICE USING SIMPLE VIBRATION SENSORS”

Author: Maria et al., 2017

This research focuses on developing a home-level earthquake alert device using basic vibration sensors. The goal was to design simple, affordable systems for common households.

Limitation: The detection accuracy is low compared to professional seismic sensors. No wireless connectivity or real-time monitoring was available.

[2] Title: “DESIGN AND IMPLEMENTATION OF AN EARTHQUAKE EARLY WARNING SYSTEM USING VIBRATION SENSORS”

Author: Sharma et al., 2018

This paper presents a low-cost earthquake alarm system using the SW-420 vibration sensor interfaced with a microcontroller. The sensor detects abnormal vibrations beyond a threshold and activates a buzzer. The study emphasizes simplicity and cost-effectiveness, making it suitable for home-level safety.

Limitation: The system does not differentiate between real seismic activity and noise vibrations (vehicles, footsteps, machine vibrations). It also lacks data logging and network-based alerting.

[3] Title: “WIRELESS SEISMIC MONITORING SYSTEM USING GSM AND ARDUINO”

Author: Fatima & Khan, 2019

This paper presents a seismic activity monitoring system with Arduino Uno and a GSM module. The focus is on sending immediate SMS alerts to authorities or building managers.

Limitation: The system generates multiple messages during continuous shaking.

[4] Title: “EARLY EARTHQUAKE DETECTION USING MEMS ACCELEROMETERS AND MICROCONTROLLERS”

Author: B. Hiroshi, 2020

This work highlights the use of MEMS accelerometers for accurate monitoring of ground acceleration. The system captures even small tremors and logs data for analysis.

Limitation: MEMS sensors increase the cost of the system. Power consumption is high, and the system requires stable power backup.

[5] Title: “ARDUINO-BASED EARTHQUAKE MONITORING SYSTEM WITH GSM ALERTING”

Author: A. Kumar & R. Patel, 2020

This paper discusses an Arduino Uno-based earthquake monitoring system using an accelerometer and GSM module. When vibrations exceed safe limits, the system sends SMS alerts to registered users. It also includes real-time data display on an LCD.

Limitation: The system depends on GSM network availability. The vibration threshold must be manually calibrated, and the sensor's accuracy reduces in noisy environments.

[6] Title: “SW-420-BASED VIBRATION MONITORING SYSTEM WITH THRESHOLD ADJUSTMENT”

Author: L. Chen & Zhou, 2021

The authors developed a vibration monitoring circuit using SW-420 with an adjustable potentiometer. The study shows how threshold tuning improves detection accuracy.

Limitation: Threshold tuning must be done manually for every environment (soft floor, concrete, upstairs rooms), making setup time-consuming.

[7] Title: “IOT-ENABLED SEISMIC ACTIVITY MONITORING SYSTEM USING CLOUD STORAGE”

Author: Mehra and Singh, 2021

This research work integrates IoT for real-time monitoring of seismic activity. It uses a NodeMCU with an accelerometer to continuously upload vibration readings to a cloud server. Users can access the data on a web dashboard.

Limitation: High power consumption and dependency on Wi-Fi make it unsuitable for rural or disaster-hit locations where power or network fails.

[8] Title: “LOW-COST EARTHQUAKE ALERT SYSTEM USING MEMS ACCELEROMETER”

Author: Tanaka et al., 2022

The authors propose a compact earthquake early warning device using a 3-axis MEMS accelerometer. The system identifies P-wave activity and triggers alarms before major shaking occurs. The design aims to reduce cost while improving responsiveness.

Limitation: Detecting P-waves accurately requires advanced signal filtering, which is computationally heavy and not suitable for basic microcontrollers like Arduino Uno without additional modules.

[9] Title: “PERFORMANCE EVALUATION OF SW-420 VIBRATION SENSOR IN HAZARD DETECTION”

Author: S. Prakash, 2023

This study evaluates the SW-420 vibration sensor specifically for detecting mechanical shocks and earthquake-like vibrations. The sensor is found to be cost-effective and highly responsive but sensitive to noise.

Limitation: High false triggering rates due to sensitivity to minor vibrations and environmental noise. The system needs calibration and filtering to improve reliability.

Chapter 3

Methodology

3.1 Block Diagram

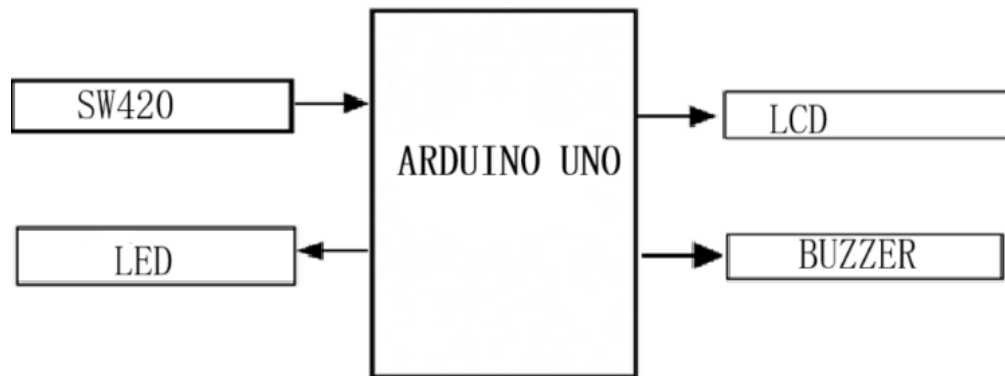


Figure 2: Earthquake monitoring and alerting system.

- **SW-420 Vibration Sensor:**

- The SW-420 is the input sensing device that detects vibrations or shocks from the environment.
- When it senses sudden movement, it changes its output signal (HIGH/LOW) and sends this data to the Arduino.
- Its built-in comparator helps filter small background vibrations.
- Not suitable for precise vibration measurement.
- Only detects presence/absence of vibration, not magnitude.

- **Arduino Uno (Microcontroller):**

- The Arduino Uno is the central processing unit of the system.
- It receives signals from the vibration sensor and checks whether they cross a predefined threshold.
- Based on the reading, it decides whether to trigger the alert system.
- It controls the output components (LCD, LED, buzzer) and manages the system workflow.

-
- Not suitable for precise vibration measurement.
 - Only detects presence/absence of vibration, not magnitude.
 - **LCD Display (16×2):**
 - The LCD screen displays important information such as:
 - Sensor values.
 - System status (“Normal”, “Vibration Detected”, “Earthquake Alert”).
 - This helps users visually understand what the system is detecting in real time.
 - The LCD is controlled by the Arduino using digital or I2C communication.
 - **LED Indicator:**
 - The LED acts as a visual alert signal.
 - When the Arduino detects high vibration levels, it turns on the LED to instantly indicate danger.
 - During normal conditions, the LED remains OFF.
 - **Buzzer:**
 - The buzzer provides an audible alarm when abnormal vibration or possible earthquake activity is detected.
 - The Arduino activates the buzzer immediately to warn people nearby.
 - It may beep continuously or in patterns depending on the code.
 - **Power Supply Block**
 - This block provides the necessary 5V DC power for the Arduino, sensor, LCD, LED, and buzzer.
 - The Arduino can be powered through USB, battery, or an adapter.
 - It ensures stable power for reliable system operation.

- **Output Alert Section:**

- This section includes the LED, buzzer, and LCD combined.
- It informs users of system status using visual alerts (LED), sound (buzzer), and messages (LCD).
- Helps ensure quick response during potential hazards.

3.2 Flow Chart

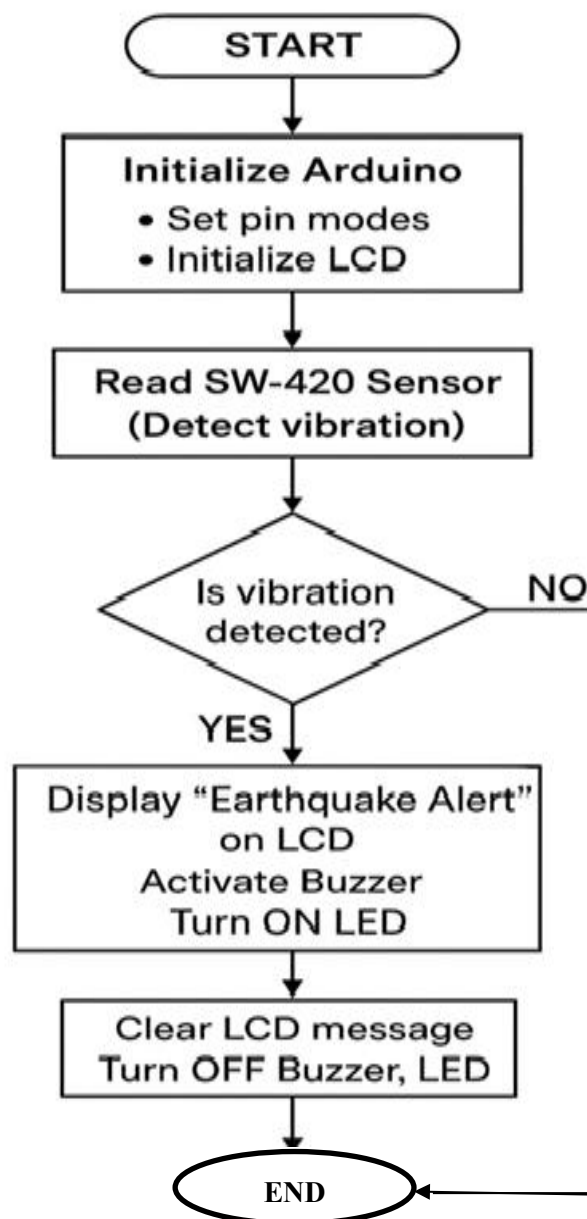


Figure 3: Flowchart of Earthquake Monitoring and Alerting System

Chapter 4

Hardware and Software Requirements

The hardware and software work together to make the earthquake monitoring system function. The hardware includes the Arduino Uno, SW-420 vibration sensor, LCD, LED, buzzer, wires, and power supply. The sensor detects vibrations, and the Arduino reads this signal and controls the LCD, LED, and buzzer to show alerts.

The software part uses the Arduino IDE, where the program is written and uploaded to the Arduino. This program tells the Arduino how to read the sensor, when to turn on the buzzer and LED, and what message to show on the LCD. Without the hardware the system cannot detect vibrations, and without the software the hardware cannot work. Both together create a simple and effective earthquake alerting system.

4.1 Hardware Requirements

- **Arduino Uno:**

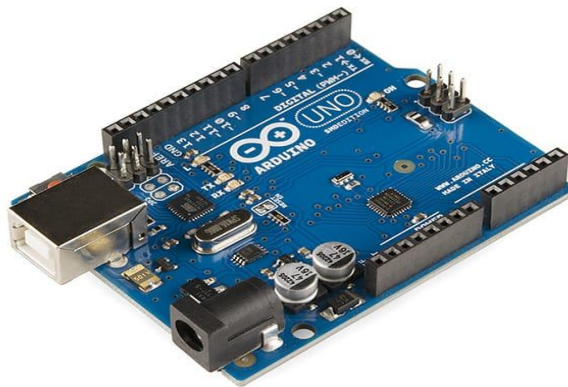


Figure 4: Arduino Uno

- Main microcontroller board for data processing and controlling all components.
- It reads input signals from the SW-420 vibration sensor, processes the data, compares it with a threshold value, and then controls the output components like the buzzer, LED, and LCD.

- **SW-420 Vibration Sensor Module:**

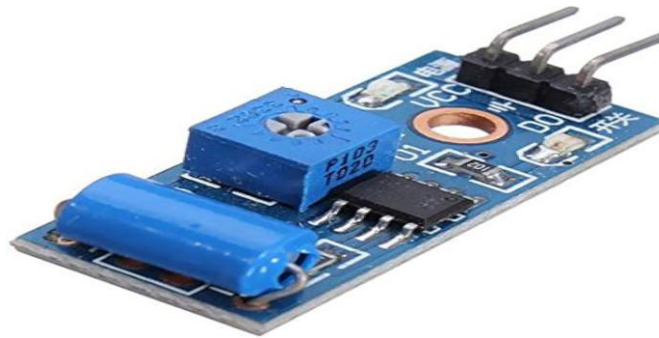


Figure 5: SW-420

- Detects vibrations or shocks and sends digital signals to Arduino.
- The sensor acts as the primary input device that identifies earthquake-like vibrations.
- **16×2 LCD Display (or I2C LCD):**

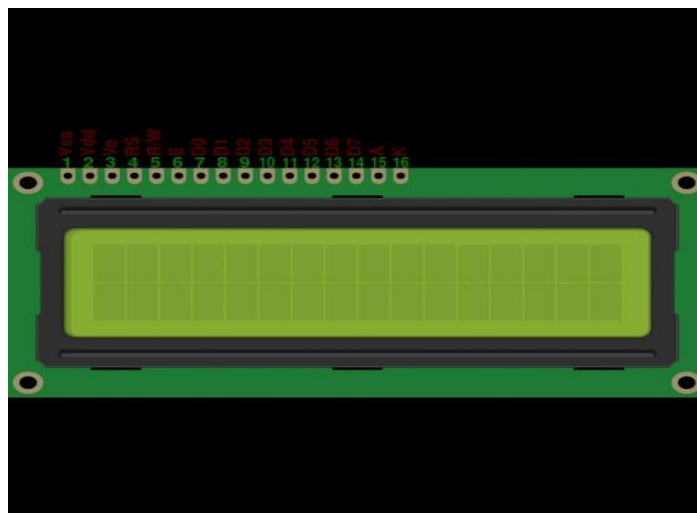


Figure 6: LCD

- The LCD is used to show the system's real-time status, such as "Normal," "Vibration Detected," or "Earthquake Alert."
- It helps the user visually understand the system response and sensor values.

- **Active Buzzer:**



Figure 7: Buzzer

- The buzzer is responsible for producing an audible alarm when abnormal vibration is detected. It warns people immediately when a possible earthquake occurs.

- **LED Indicator:**



Figure 7: LED

- LED Provides visual alert when vibration exceeds threshold.
 - The LED provides a quick visual indication of the alert condition. It glows whenever the system detects vibration above the threshold.
- **Resistors (220Ω for LED):**
 - These resistors are used to limit the current going into the LED, preventing damage and ensuring safe operation.
 - **Breadboard:**
 - A breadboard is used for temporary circuit connections. It helps assemble all components neatly without soldering.

- **Jumper Wires:**

- Jumper wires allow electrical connections between the Arduino, sensor, LCD, LED, and buzzer on the breadboard.

- **Power Supply:**

- The Arduino and all modules operate on 5V DC. Power can be supplied through USB, battery, or an adapter, ensuring stable and reliable operation.

- **Optional Components:**

- Potentiometer – to adjust LCD contrast.
- Push button – to reset or stop the alarm.
- Transistor + Diode – used if powering high-current buzzers.
- SD Card Module – for logging vibration data.

4.2 Software Requirements

- **Arduino IDE:**



Figure 8: Arduino IDE Software

- The Arduino Integrated Development Environment is used to write, compile, and upload the program to the Arduino Uno. It also provides tools like the Serial Monitor to debug sensor readings and test system behavior.

- **Arduino Libraries:**

- Certain libraries make coding easier and allow smooth communication between components.

Examples:

- LiquidCrystal.h – for controlling the standard 16×2 LCD.
- LiquidCrystal_I2C.h – for I2C LCD displays.
- Wire.h – for handling I2C communication.
- These libraries simplify displaying text, reading sensors, and sending commands.

- **Computer or Laptop:**

- A computer is necessary to write the Arduino code, upload it to the UNO, and run tests. It serves as the development platform for the entire project.

- **USB Drivers:**

- When the Arduino is connected to the computer, USB drivers allow it to communicate properly. These drivers are usually installed automatically with the Arduino IDE.

Chapter 5

Results and Discussions

The earthquake monitoring and alerting system successfully detected vibrations using the SW-420 sensor and provided instant alerts through the buzzer, LED, and LCD display. During testing, the system showed quick response and stable performance, accurately identifying vibration levels above the set threshold. The LCD displayed clear status messages such as “Normal,” “Vibration Detected,” or “Earthquake Alert,” which helped users monitor the system in real time. The buzzer produced a loud alarm and the LED glowed immediately when strong vibrations were detected, ensuring fast user awareness. The system also proved energy-efficient and operated smoothly over long durations without errors. However, the sensor sometimes responded to minor non-seismic vibrations like table movements, which shows the need for proper calibration. Overall, the project demonstrated that a simple, low-cost Arduino-based system can effectively monitor vibrations and provide a basic early warning signal for small-scale environments.

The image shows a basic earthquake monitoring and alerting system built using an Arduino Uno. A vibration sensor detects any shaking, and the LCD screen displays the status, such as “No vibration.” An LED and a buzzer are connected to give visual and sound alerts when vibrations are detected. The setup uses simple wiring on a breadboard to show how low-cost components can be combined to create a small earthquake alert system.

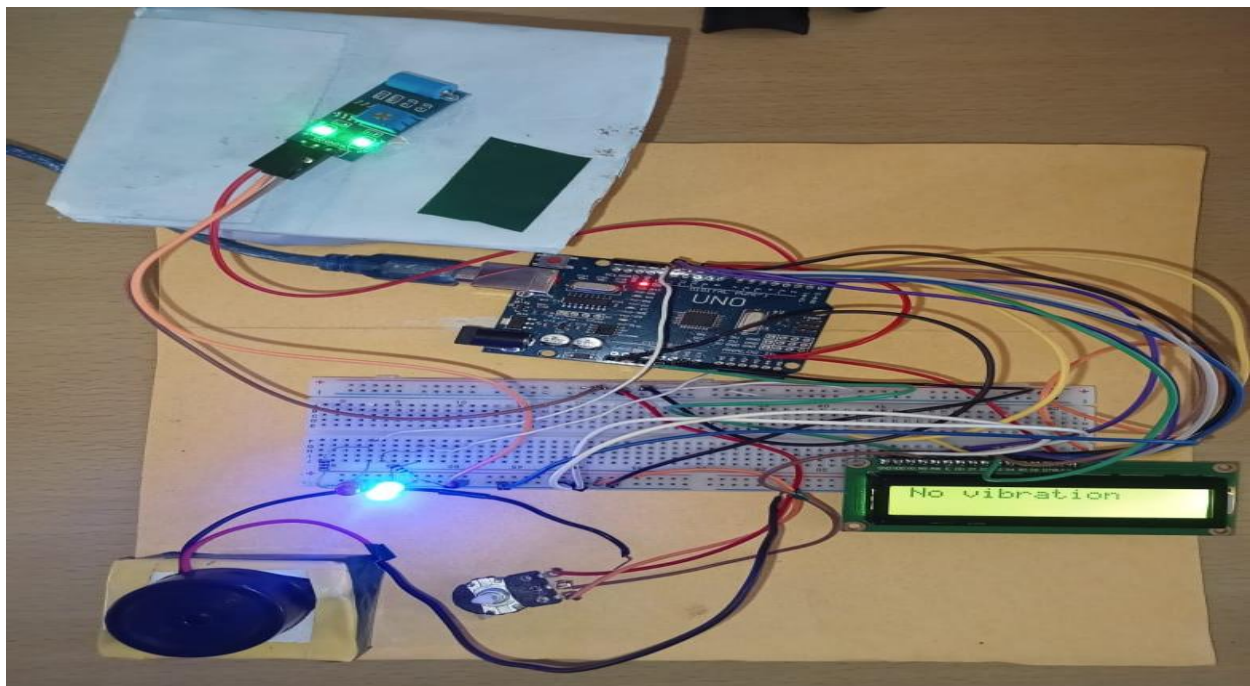


Figure 9: Before Earthquake Monitoring and Alerting System

The image shows the earthquake monitoring and alerting system in its alert state. When the vibration sensor detects shaking, the system activates the red LED and the buzzer, indicating an earthquake. The LCD screen displays the message “Earthquake!!” to show that vibration has been detected. The Arduino Uno controls all components, and the wiring on the breadboard connects the sensor, buzzer, LED, and display to create the alerting system.

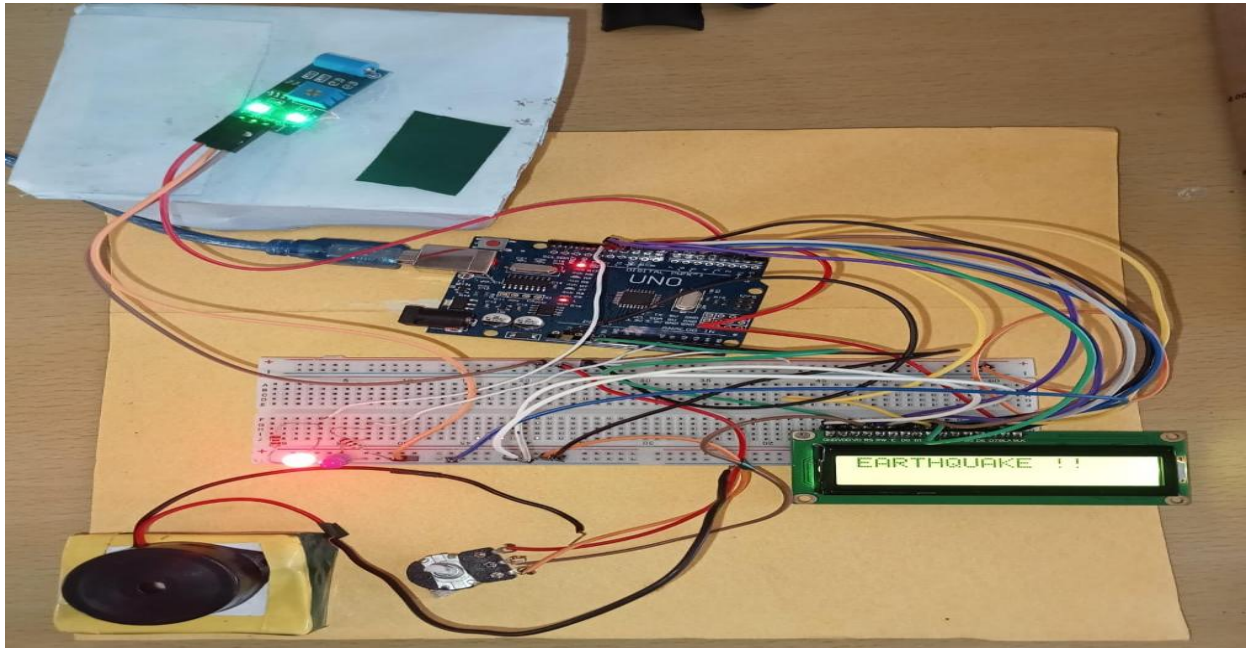


Figure 10: After Earthquake Monitoring and Alerting System

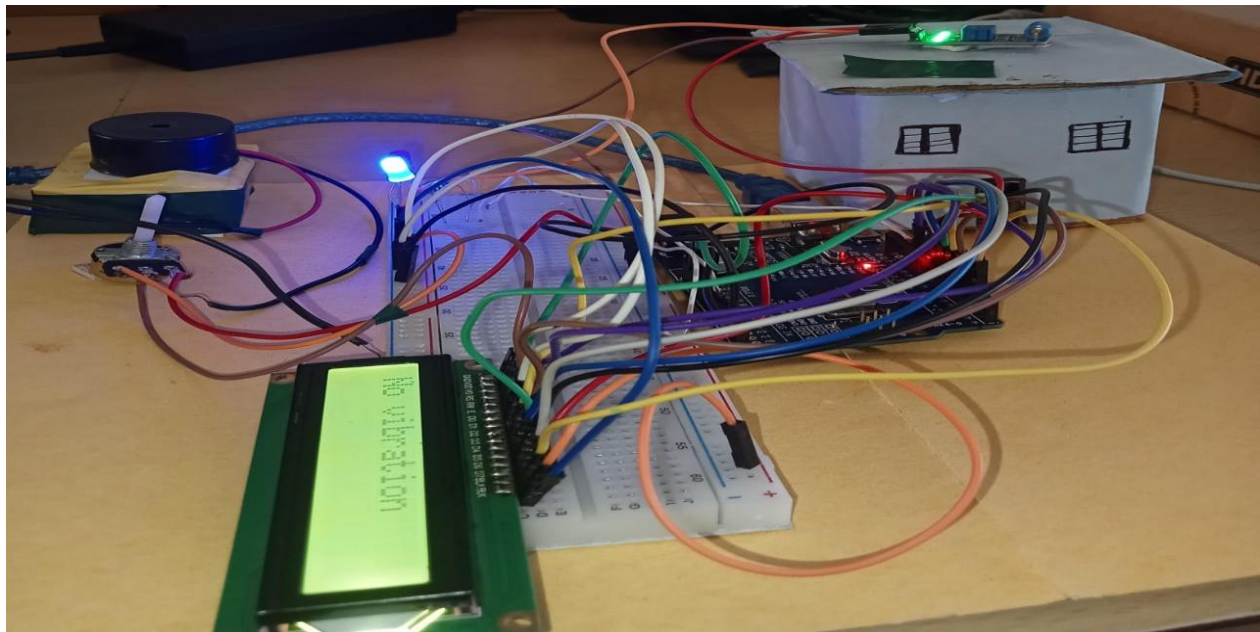


Figure 11: Final output

Chapter 6

Advantages, Disadvantages and Applications

6.1 Advantages

- **Low Cost:**
 - Uses inexpensive components like Arduino Uno and SW-420 sensor, making the system affordable.
- **Easy to Build:**
 - Requires simple wiring and basic coding; suitable for students and beginners.
- **Quick Response:**
 - The system detects vibrations instantly and activates alerts within milliseconds.
- **Real-Time Monitoring:**
 - The LCD continuously displays system status: normal, vibration, or earthquake alert.
- **Dual Alert System:**
 - Both LED and buzzer provide visual and audible warnings, improving safety.
- **Low Power Consumption:**
 - Operates on 5V, can run on USB or battery, making it energy efficient.
- **Portable System:**
 - Light-weight and compact; can be placed anywhere indoors.
- **Reliable for Small-Scale Safety:**
 - Works well in homes, classrooms, laboratories, offices, and small buildings.
- **Simple Maintenance:**
 - Requires minimal maintenance; easy to test and calibrate.
- **Good for Learning:**
 - Helps students understand concepts of sensors, microcontrollers, and real-time systems.

- **Immediate Feedback:**
 - Alerts the user instantly when abnormal vibration occurs, helping reduce delay.
- **Customizable:**
 - More features can be added like GSM alerts, IoT monitoring, or advanced sensors.

6.2 Disadvantages/Limitations

- **Limited Accuracy:**
 - The SW-420 sensor only detects vibrations; it cannot measure actual earthquake magnitude or intensity.
- **False Alarms:**
 - Sensor may trigger due to normal movements like door closing, people walking, vehicles passing, or table shaking.
- **No Distance Detection:**
 - The system cannot detect how far or how near the earthquake source is.
- **Not Suitable for Large Buildings:**
 - Works only for small rooms or indoor areas; cannot protect large structures or cities.
- **No Data Logging:**
 - The system does not store vibration data for analysis unless extra modules are added.
- **Limited Sensor Capability:**
 - SW-420 cannot differentiate between small harmless vibrations and real seismic activity without careful calibration.
- **No Remote Alerts:**
 - The basic system cannot send SMS, mobile notifications, or internet alerts.
- **Environmental Sensitivity:**
 - Temperature, humidity, and surface placement can affect sensor accuracy.

6.3 Applications

- **Home Safety System:**
 - Can be installed in houses to give early warnings of sudden vibrations or tremors.
- **School and College Labs:**
 - Useful in electronics, IoT, and engineering labs for demonstrating vibration detection and alert systems.
- **Industrial Machinery Monitoring:**
 - Can detect abnormal vibrations in machines to prevent damage or accidents.
- **Offices and Small Buildings:**
 - Provides basic vibration alerts for small workplaces.
- **Educational Projects:**
 - Excellent for students learning sensors, microcontrollers, and real-time monitoring.
- **Research and Prototyping:**
 - Helps in testing vibration behavior for small-scale seismic studies or engineering experiments.
- **Safety Demonstrations:**
 - Useful for teaching natural disaster awareness and safety procedures.
- **Laboratory Equipment Protection:**
 - Can be placed near sensitive instruments to detect unwanted vibrations.
- **Temporary Monitoring in Remote Areas:**
 - Lightweight and portable; can be quickly deployed in remote or temporary monitoring locations.

Chapter 7

Conclusion and Future Scope

The earthquake monitoring and alerting system provides a low-cost, efficient solution for detecting ground vibrations and sending timely alerts through alarms and SMS messages. Using components like Arduino, SW-420 vibration sensor, buzzer, LED, and LCD display, the system can monitor seismic activity in real time and improve safety in homes, schools, offices, and public places. While it may not match the precision of professional seismic equipment, it serves as a practical tool for early warning and disaster preparedness.

7.1 Conclusion

The Earthquake Monitoring and Alerting System designed using Arduino Uno, the SW-420 vibration sensor, a buzzer, LED, and LCD display successfully demonstrates the ability of low-cost embedded systems to detect vibrations and provide timely alerts. The project proves that even with basic sensors, it is possible to build an effective early warning mechanism for small-scale environments. Throughout testing, the system showed quick response, stable performance, and clear status indication, making it suitable for indoor monitoring. The project highlights the importance of real-time processing, as the Arduino continuously monitors sensor input and activates alerts without delay. The LCD enhances user awareness by displaying system status, while the buzzer and LED ensure immediate attention during abnormal vibrations. Although the SW-420 sensor has limitations in accuracy and may produce false alarms, proper calibration greatly improves its reliability. Overall, the system provides a strong foundation for understanding vibration detection and disaster warning technologies. It can be enhanced further with advanced sensors, wireless communication modules, and IoT integration to create a more accurate and scalable earthquake alert system. Thus, the project achieves its objective of showcasing a simple, affordable, and practical approach to early vibration detection and contributes to increasing safety awareness and preparedness.

7.2 Future Scope

In the coming years, the earthquake monitoring and alerting system can be enhanced in several advanced ways to make it more accurate, intelligent, and widely applicable. Instead of using a basic vibration sensor, the system can be upgraded with high-precision accelerometers, gyroscopes, or geophone sensors to measure ground movement more accurately and reduce false alarms. The entire setup can be connected to the Internet of Things (IoT) so that vibration data is uploaded to cloud dashboards in real time, allowing users, researchers, and authorities to monitor seismic activity from any where. The system can also be expanded into a multi-sensor network across multiple rooms, buildings, or even a small locality to create a mini-earthquake early warning system. Adding data logging and machine learning can help predict unusual vibration patterns and improve system intelligence over time. Solar panels or battery backup can ensure uninterrupted operation even during power failures, which is crucial in disaster situations. In the future, this system can also integrate with smart home devices, fire alarms, or automated safety systems, making it a more reliable, scalable, and impactful solution for community safety and disaster preparedness.

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